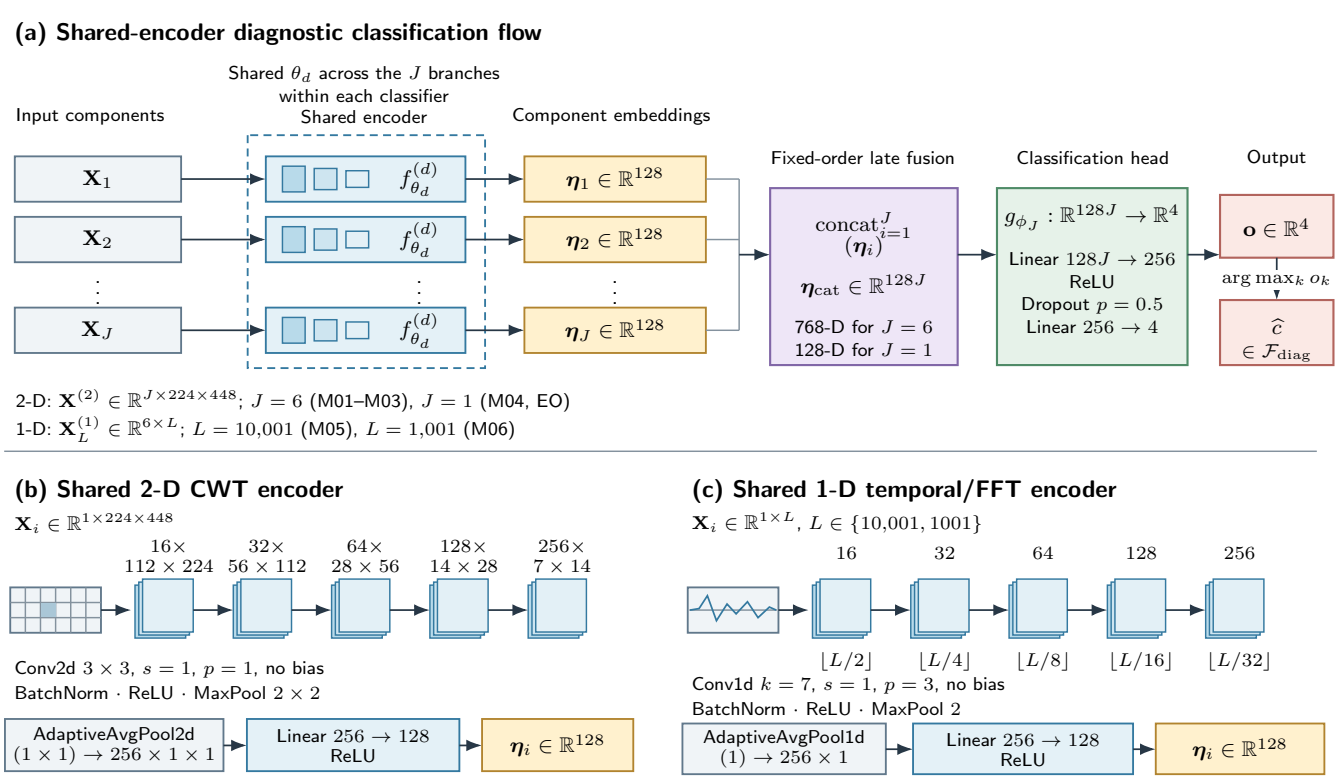

(a) Shared-encoder diagnostic classification flow
Shared θ_d across the J branches within each classifier
Shared encoder
Input components
Component embeddings
$\mathbf{X}_1$
$\mathbf{X}_2$
$\mathbf{X}_J$
$f^{(d)}_{\theta_d}$
$\boldsymbol{\eta}_1 \in \mathbb{R}^{128}$
$\boldsymbol{\eta}_2 \in \mathbb{R}^{128}$
$\boldsymbol{\eta}_J \in \mathbb{R}^{128}$
Fixed-order late fusion
$\mathrm{concat}_{i=1}^{J}(\boldsymbol{\eta}_i)$
$\boldsymbol{\eta}_{\mathrm{cat}} \in \mathbb{R}^{128J}$
768-D for $J = 6$
128-D for $J = 1$
Classification head
$g_{\phi_J} : \mathbb{R}^{128J} \to \mathbb{R}^4$
Linear $128J \to 256$
ReLU
Dropout $p = 0.5$
Linear $256 \to 4$
Output
$\mathbf{o} \in \mathbb{R}^4$
$\arg\max_k o_k$
$\widehat{c} \in \mathcal{F}_{\mathrm{diag}}$
2-D: $\mathbf{X}^{(2)} \in \mathbb{R}^{J\times 224\times 448}$; $J = 6$ (M01–M03), $J = 1$ (M04, EO)
1-D: $\mathbf{X}_L^{(1)} \in \mathbb{R}^{6\times L}$; $L = 10{,}001$ (M05), $L = 1{,}001$ (M06)
(b) Shared 2-D CWT encoder
$\mathbf{X}_i \in \mathbb{R}^{1\times 224\times 448}$
16× 112 × 224
32× 56 × 112
64× 28 × 56
128× 14 × 28
256× 7 × 14
Conv2d 3×3, $s = 1$, $p = 1$, no bias
BatchNorm · ReLU · MaxPool 2×2
AdaptiveAvgPool2d $(1 \times 1) \to 256 \times 1 \times 1$
Linear $256 \to 128$
ReLU
$\boldsymbol{\eta}_i \in \mathbb{R}^{128}$
(c) Shared 1-D temporal/FFT encoder
$\mathbf{X}_i \in \mathbb{R}^{1\times L}$, $L \in \{10{,}001, 1001\}$
16
32
64
128
256
$\lfloor L/2 \rfloor$
$\lfloor L/4 \rfloor$
$\lfloor L/8 \rfloor$
$\lfloor L/16 \rfloor$
$\lfloor L/32 \rfloor$
Conv1d $k = 7$, $s = 1$, $p = 3$, no bias
BatchNorm · ReLU · MaxPool 2
AdaptiveAvgPool1d $(1) \to 256 \times 1$
Linear $256 \to 128$
ReLU
$\boldsymbol{\eta}_i \in \mathbb{R}^{128}$